

Early disease detection: IoT devices continuously monitor and analyse patient data in real time, enabling the early detection of signs of diseases or health deterioration. This leads to timely interventions and improved health outcomes.

Efficient hospital operations: IoT optimises hospital operations by tracking equipment, managing inventory, and monitoring environmental conditions. This optimisation leads to cost savings, improved efficiency, and enhanced patient safety.

Open source simulation tools for IoT implementation

Several open source simulation tools can be used for implementing and testing IoT applications. These tools offer flexibility and customisation,

making them suitable for many IoT projects. Table 1 gives a comparison of popular IoT simulation tools.

Python libraries for IoT-enabled healthcare implementation

Python is a versatile programming language with many libraries that can be used in IoT-enabled healthcare implementations. Table 2 lists popular

Python libraries that are commonly used in IoT-enabled healthcare projects.

IoT is not just changing how we think about healthcare — it's changing how we experience it. Patients are more engaged, doctors better informed, and hospitals vastly more efficient. The Internet of Things is the connective thread, creating a more inter-connected, efficient, and compassionate healthcare system. **END** 

References

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- **Synergy with deep learning:** As deep learning models become more intricate, their computational demands skyrocket. Neuromorphic systems, with their inherent efficiency and brain-like processing, could become the preferred substrate for running these complex models, especially in power-constrained IoT environments.

Emergence of quantum computing

- **Beyond classical computing:** Quantum computing represents a leap beyond classical computing principles, offering exponentially faster computations for specific tasks. However, the intersection of quantum and neuromorphic computing remains largely uncharted, promising unprecedented computational capabilities.
- **Potential synergy:** While neuromorphic systems emulate the brain's structure and functionality,

quantum systems could potentially capture its underlying quantum mechanical properties. This synergy could unlock new models of neural processing, further enhancing IoT capabilities.

- **IoT and quantum:** As quantum technologies mature, they'll demand architectures that can seamlessly integrate with them. Neuromorphic systems, with their adaptability and bio-inspired designs, may offer the most compatible bridge to this quantum future.

In the confluence of AI and quantum computing, neuromorphic systems stand as potent allies. Their

brain-inspired designs resonate with AI's aspirations, and their adaptive architectures might dovetail with quantum's complexities. As IoT devices become ubiquitous, smarter, and more integrated into our lives, neuromorphic computing's relevance will only amplify, heralding a future where devices don't just connect but also cogitate.

Neuromorphic computing is more than just another tech buzzword; it presents a genuine paradigm shift. By drawing inspiration from the most complex structure known to us, the human brain, we're not merely aiming for better computing; we're redefining what it means to compute. **END** 

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The author is a research analyst with a specialisation in artificial intelligence and machine learning. He is the founder of Arkinfo, which focuses on the research and development of tech products using new age technologies. He shares a deep love for the analysis of technological trends and understanding their implications. Being a FOSS enthusiast, he has contributed to several open source projects.